

MODULE-9:

Sure, here are some important points from the AWS Academy Cloud Foundations Module 9 Student Guide, making sure not to miss anything:

- **The AWS Well-Architected Framework**

- The AWS Well-Architected Framework is a guide designed to help build secure, high-performing, resilient, and efficient infrastructures for applications and workloads in the cloud.
- The framework is based on six pillars: operational excellence, security, reliability, performance efficiency, cost optimization, and sustainability.
 - **Operational excellence** focuses on running and monitoring systems to deliver business value and continually improving processes and procedures.
 - **Security** emphasizes protecting information, systems, and assets while delivering business value through risk assessments and mitigation strategies.
 - **Reliability** ensures that a workload performs its intended function correctly and consistently when expected.
 - **Performance efficiency** is about using IT and computing resources efficiently to meet system requirements and maintain efficiency as demand changes and technologies evolve.
 - **Cost optimization** focuses on avoiding unnecessary costs.
 - **Sustainability** was added in 2021 to help organizations learn how to minimize the environmental impacts of running cloud workloads.
- Each pillar includes a set of design principles and best practice areas. Each best practice area aligns to questions that a reviewer should ask when designing an architecture.
- The document focuses on the first five pillars and leads you through a review of an example architecture against each pillar's design principles. For more about the sustainability pillar, refer to the sustainability pillar section within the Well-Architected Framework documentation.

- **Reliability and Availability**

- **Reliability** is a measure of a system's ability to provide functionality when desired by the user.
 - It includes all system components: hardware, firmware, and software.
 - It's the probability that your entire system will function as intended for a specified period.
 - Mean time between failures (MTBF) = total time in service/number of failures
- **Availability** is the percentage of time that a system is operating normally.
 - It can also be defined as the percentage of uptime over a period of time.
 - It is often expressed as "number of 9s." For example, five 9s means 99.999 percent availability.
- **High availability** means that a system can withstand some measure of degradation while remaining available.

- Downtime is minimized.
 - Minimal human intervention is required.
- **Factors that influence availability:**
 - **Fault tolerance:** the built-in redundancy of an application's components and its ability to remain operational.
 - **Scalability:** the ability of an application to accommodate increases in capacity needs without changing design.
 - **Recoverability:** the process, policies, and procedures related to restoring service after a catastrophic event.
- **AWS Trusted Advisor**
 - AWS Trusted Advisor is an online tool that provides real-time guidance to help you provision resources following AWS best practices.
 - It looks at your entire AWS environment and gives you real-time recommendations in five categories: cost optimization, performance, security, fault tolerance, and service limits.
 - **Cost Optimization:** AWS Trusted Advisor looks at your resource use and makes recommendations to help you optimize cost by eliminating unused and idle resources, or by making commitments to reserved capacity.
 - **Performance:** Improve the performance of your service by checking your service limits, ensuring you take advantage of provisioned throughput, and monitoring for overutilized instances.
 - **Security:** Improve the security of your application by closing gaps, enabling various AWS security features, and examining your permissions.
 - **Fault Tolerance:** Increase the availability and redundancy of your AWS application by taking advantage of automatic scaling, health checks, Multi-AZ deployments, and backup capabilities.
 - **Service Limits:** AWS Trusted Advisor checks for service usage that is more than 80 percent of the service limit. Values are based on a snapshot, so your current usage might differ. Limit and usage data can take up to 24 hours to reflect any changes.

Let me know if you need further information.